

# DSA Day 32: Graph Representations

Book: Data Structures Through C in Depth - Ch7 §7.10-7.11 · pp 335-352 | Code in Python

## Overview

Adjacency matrix ( $O(V^2)$  space,  $O(1)$  edge test) vs adjacency list ( $O(V+E)$  space). Warshall computes transitive closure.

## Key concepts to master

- matrix vs list trade-offs
- transitive closure (Warshall)

## Python implementation

```
def warshall(m):
    n=len(m)
    for k in range(n):
        for i in range(n):
            for j in range(n):
                m[i][j]=m[i][j] or (m[i][k] and m[k][j])
    return m
```

## Complexity

Matrix  $O(V^2)$ ; list  $O(V+E)$ ; Warshall  $O(V^3)$ .

## Common pitfalls

- Using a matrix for a sparse graph
- Off-by-one in Warshall loops

## Practice

- Store a 5-node graph both ways
- Compute its path matrix

